

Disproof of TLMC Conjecture 00000000544: the Equality-on-Forests Clause Fails on K_2

TLMC submission — conjecture 00000000544

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1 The conjecture

Let G be a graph without isolated vertices, A_G its incidence matrix, and $I_{A_G} \subseteq k[x_1, \dots, x_{|E|}]$ the toric ideal generated by A_G , i.e. the kernel of the monomial map

$$\varphi : k[x_1, \dots, x_{|E|}] \longrightarrow k[t_1^{\pm 1}, \dots, t_{|V|}^{\pm 1}], \quad x_e \longmapsto t^{a_e},$$

where a_e is the column of A_G indexed by the edge e . The Graver basis $\text{Graver}(I_{A_G})$ is the set of all irreducible elements of the lattice $L_A = \ker_{\mathbb{Z}}(A) \subseteq \mathbb{Z}^{|E|}$.

Conjecture **00000000544** asserts that for all graphs without isolated vertices

$$|\text{Graver}(I_{A_G})| \leq \sum_{H \subseteq G, H \text{ forest}} 2^{|V(H)|}, \quad \text{with equality attained on forests.} \quad (1)$$

2 The counterexample

Claim 1. $G = K_2$ (one edge, two vertices) satisfies the hypothesis but violates the equality clause of (1). The conjecture is false.

K_2 is a forest and has no isolated vertices, so (1) must hold with equality for $G = K_2$. We compute both sides.

2.1 Left-hand side: $|\text{Graver}(I_{A_{K_2}})| = 0$

The (unoriented) incidence matrix of K_2 is the 2×1 matrix

$$A_{K_2} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The monomial map is

$$\varphi : k[x_1] \rightarrow k[t_1, t_2], \quad x_1 \mapsto t_1 t_2,$$

which is injective: $t_1 t_2$ is a nonconstant monomial, so $\ker \varphi = 0$ and $I_{A_{K_2}} = 0$. (Under the oriented convention $a_1 = (1, -1)^T$ one gets $x_1 \mapsto t_1 t_2^{-1}$, again injective; nothing below changes.)

Equivalently at the lattice level,

$$L_A = \ker_{\mathbb{Z}}(A_{K_2}) = \{u \in \mathbb{Z}^1 : A_{K_2} u = 0\} = \{u \in \mathbb{Z} : u = 0\} = \{0\},$$

since $A_{K_2} u = (u, u)^T$. The lattice $\{0\}$ has no nonzero elements and therefore no irreducible ones:

$$\text{Graver}(I_{A_{K_2}}) = \emptyset, \quad |\text{Graver}(I_{A_{K_2}})| = 0.$$

2.2 Right-hand side: $\text{RHS} \geq 4 > 0$

The graph $H = G = K_2$ is itself a forest subgraph of G with $|V(H)| = 2$, so its summand alone contributes

$$2^{|V(H)|} = 2^2 = 4.$$

Hence the right-hand side of (1) is at least 4 under every reading of “ $H \subseteq G$ ”. For instance, with H ranging over spanning subgraphs there are exactly two of them, the edgeless graph and K_2 itself, both forests on 2 vertices, giving $\text{RHS} = 2^2 + 2^2 = 8$.

2.3 Conclusion

$$0 = |\text{Graver}(I_{A_{K_2}})| \neq \text{RHS} \geq 4,$$

so (1) fails with equality on the forest K_2 . Conjecture 00000000544 is **false**.

Remark 1 (robustness). *The failure is not an artefact of the smallest example. For every nonempty forest G , the columns of the unoriented incidence matrix A_G are linearly independent over $\mathbb{Q}$, hence $\ker_{\mathbb{Z}}(A_G) = \{0\}$, $I_{A_G} = 0$ and $|\text{Graver}(I_{A_G})| = 0$, while the right-hand side is $\geq 2^2 = 4$ as soon as G has an edge. Thus equality fails on every forest with at least one edge; K_2 is the minimal counterexample, and the inequality half of (1) survives here only trivially as $0 \leq \text{RHS}$.*

Remark 2 (boundary). *Single-vertex graphs are excluded by the no-isolated-vertices hypothesis, so K_2 is the smallest admissible graph; there is no smaller boundary case. The verification is machine-checked: a standalone script (**reproduce.py**) recomputes the lattice, the Graver basis and the subgraph sum, and a Lean 4 formalization (**lean4/**, core only, zero axioms, zero **sorry**) certifies all numerical assertions.*